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Constant Propagation:

This one, similarly to available expressions, was mostly filled out for us with some TODO's marked to add code in. For the first one we added functionality to the binary evaluations. This one was quite simple, we first added a check to see if either operand was a bottom and if so returned bottom as that should persist and not allow a bottom to go back to being a top. Then we added functionality for the rest of the binary operators to return the proper constant based on their evaluation.

Next we added to the phi evaluation, and were able to utilize some of the already implemented functionality for the pass to make things a little easier. When checking all the predecessors of the phi we can only fold the value if all routes lead to the phi value being the same constant value. To do this we can iterate through all the predecessors and convert them to the LVal using the evalValue function and then use it with the meetVal function to get the meet of that and an arbitrary final value that we set equal to the meetVal function. So as we iterate we're meeting each value with basically the output of the last. Then after checking all of them we also needed to add a check if the phi node itself was a predecessor and do the meet function with that or it can get caught in a loop where the value is constantly changing.

The last part was a section of the transfer function that actually didn't need any additions as any other operators either don't return a value or aren't considered in our example so we didn't add anything to the functionality there.